



Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	2d, double precision, pressure-based, laminar, transient
Version	25.2.0-10204
Source Revision	5eecd5d865
Build Time	Jun 16 2025 10:44:41 EDT
CPU	AMD Ryzen 7 6800H with Radeon Graphics
OS	Windows

Geometry and Mesh

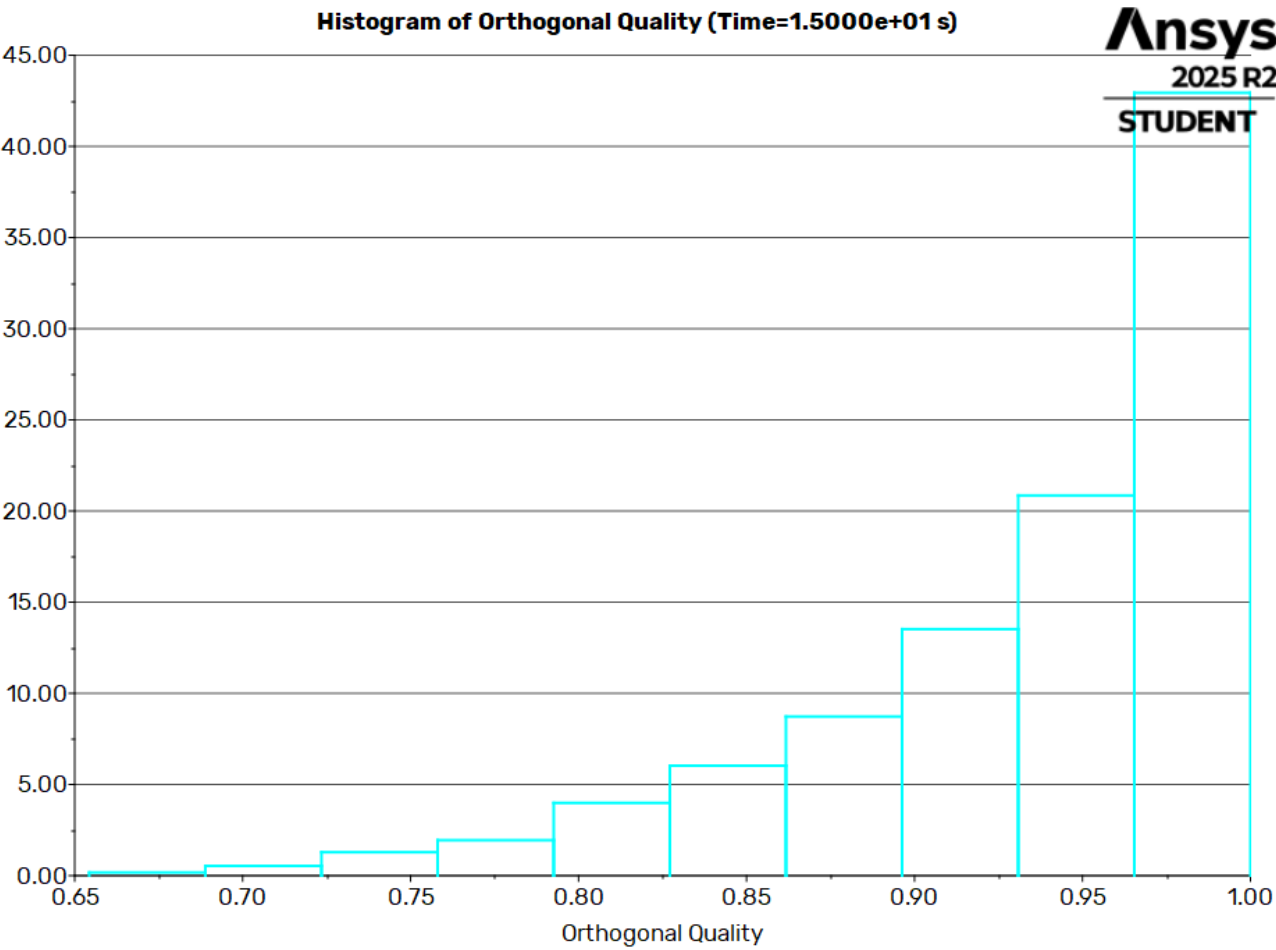
Mesh Size

Cells	Faces	Nodes
5958	8997	3040

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Tri Cell	0.65433157	3.7971083
fluid_source	Tri Cell	0.65707038	3.7953515

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Unsteady, 1st-Order Implicit
Viscous	Laminar

Material Properties

– Fluid	
– water-liquid	
Density	998.2 kg/m ³
Viscosity	0.001003 kg/(m s)
– air	
Density	1.225 kg/m ³
Viscosity	1.7894e-05 kg/(m s)
– Solid	
– aluminum	
Density	2719 kg/m ³

Cell Zone Conditions

– Fluid	
– fluid	
Material Name	water liquid
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no

Mesh Motion?	no
Porous zone?	no
– fluid_source	
Material Name	water liquid
Specify source terms?	yes
– Source Terms	
mass	mass_source::libudflnward
x-momentum	x_mom_source::libudflnward
y-momentum	y_mom_source::libudflnward
Specify fixed values?	no
Frame Motion?	no
Mesh Motion?	no
Porous zone?	no

Boundary Conditions

– Inlet	
– inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute
Velocity Magnitude	Inlet_Velocity
Supersonic/Initial Gauge Pressure [Pa]	0
– Outlet	
– outlet	
Backflow Reference Frame	Absolute
Gauge Pressure	Outlet_GaugePressure
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no

– Wall	
– walls	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

Reference Values

Area	1 m ²
Density	1.225 kg/m ³
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fluid_source

Solver Settings

– Equations	
Flow	True
– Numerics	
Absolute Velocity Formulation	True
– Unsteady Calculation Parameters	
Number of Time Steps	150
Time Step Size [s]	0.1
Max Iterations/Time Step	50
– Under-Relaxation Factors	
Pressure	0.3
Density	1
Body Forces	1

Momentum	0.7
– Pressure-Velocity Coupling	
Type	SIMPLE
– Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
– Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	2
Case Read	10.986 seconds
Iteration	19.087 seconds
AMG	6.306 seconds
Virtual Current Memory	1.00473 GB
Virtual Peak Memory	1.01178 GB
Memory Per M Cell	60.3795

Solution Status

Flow Time: 15

Time Step: 150

	Value	Absolute Criteria	Convergence Status
continuity	0.0009325448	0.001	Converged
x-velocity	1.962833e-06	0.001	Converged
y-velocity	3.891566e-08	0.001	Converged

Named Expressions

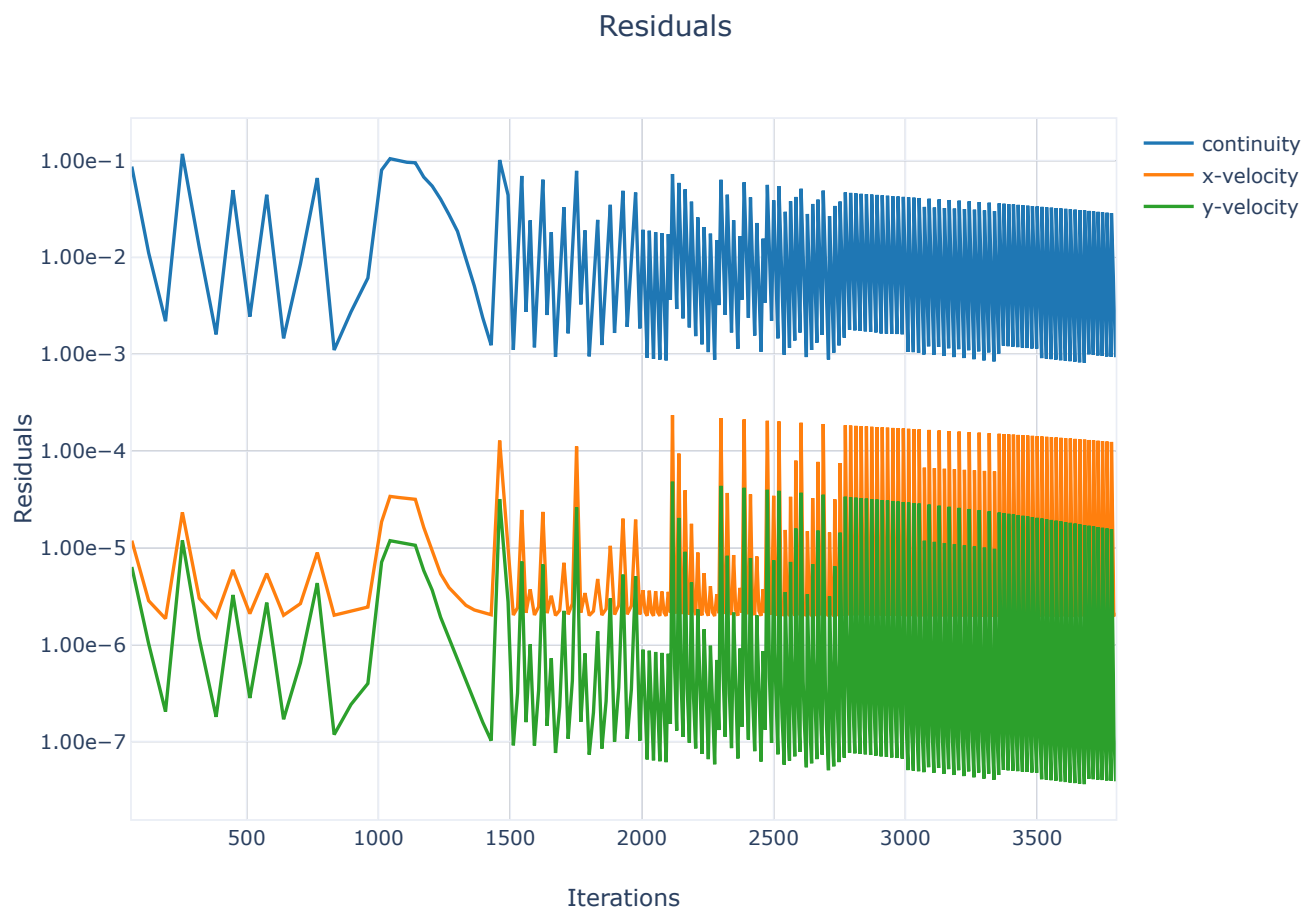
Expression	Definition	Value	Unit	Used In	Description
Inlet_Velocity	0.01 [m/s]	0.01	[m s ⁻¹]	inlet (Velocity Magnitude)	
Outlet_GaugePressure	0 [Pa]	0	[kg m ⁻¹ s ⁻²]	outlet (Gauge Pressure)	
massFlow	-10 [kg/(m ³ s)]	-10	[kg m ⁻³ s ⁻¹]		
xMomentum	10 [N/m ³]	10	[kg m ⁻² s ⁻²]		
yMomentum	10 [N/m ³]	10	[kg m ⁻² s ⁻²]		

Report Definitions

delta-time	0.1	s
iters-per-timestep	17	
flow-time	15	s

Plots

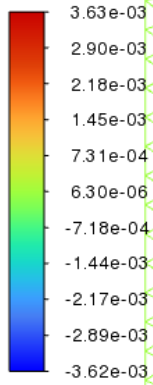
Residuals



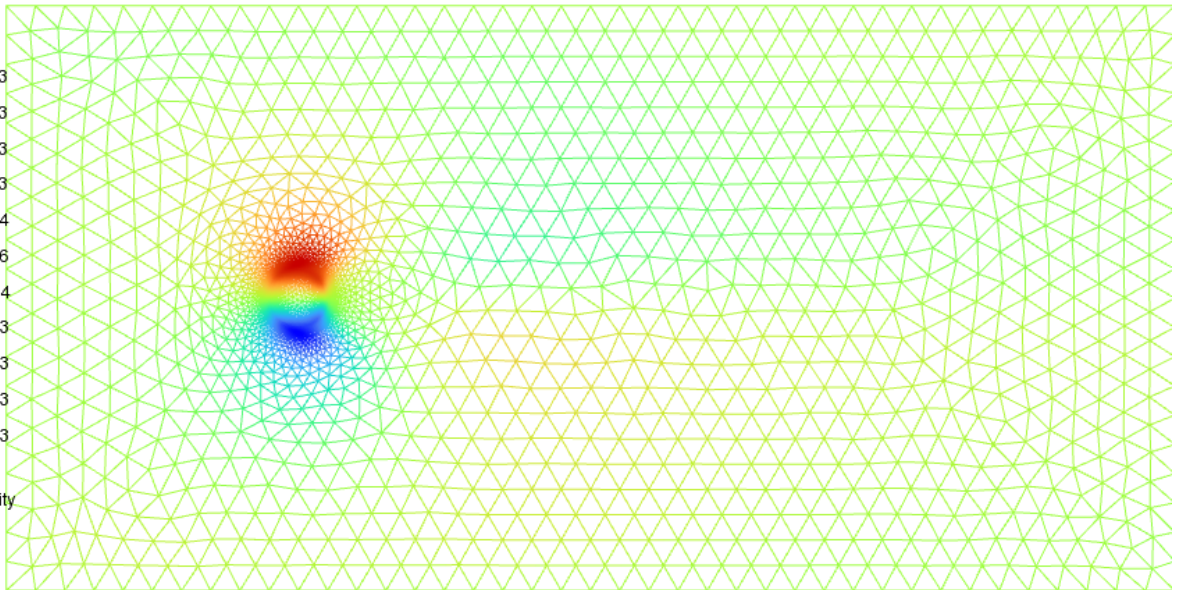
Contours

contour-1-y-velocity

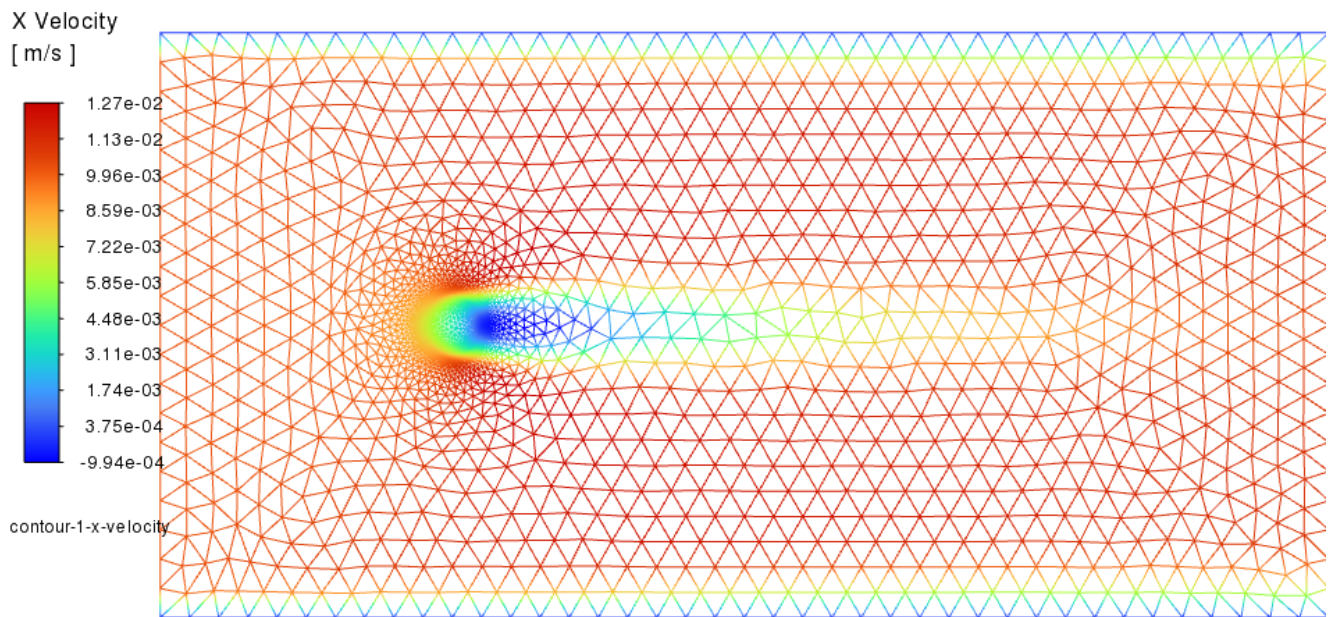
Y Velocity
[m/s]



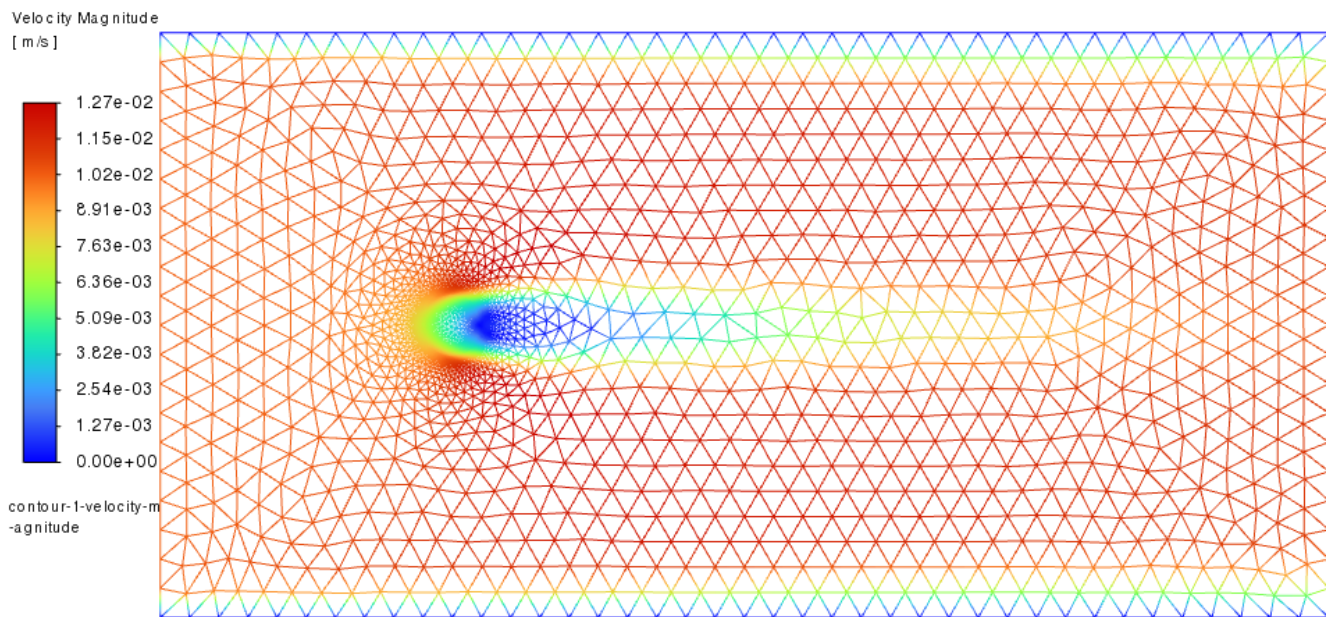
contour-1-y-velocity



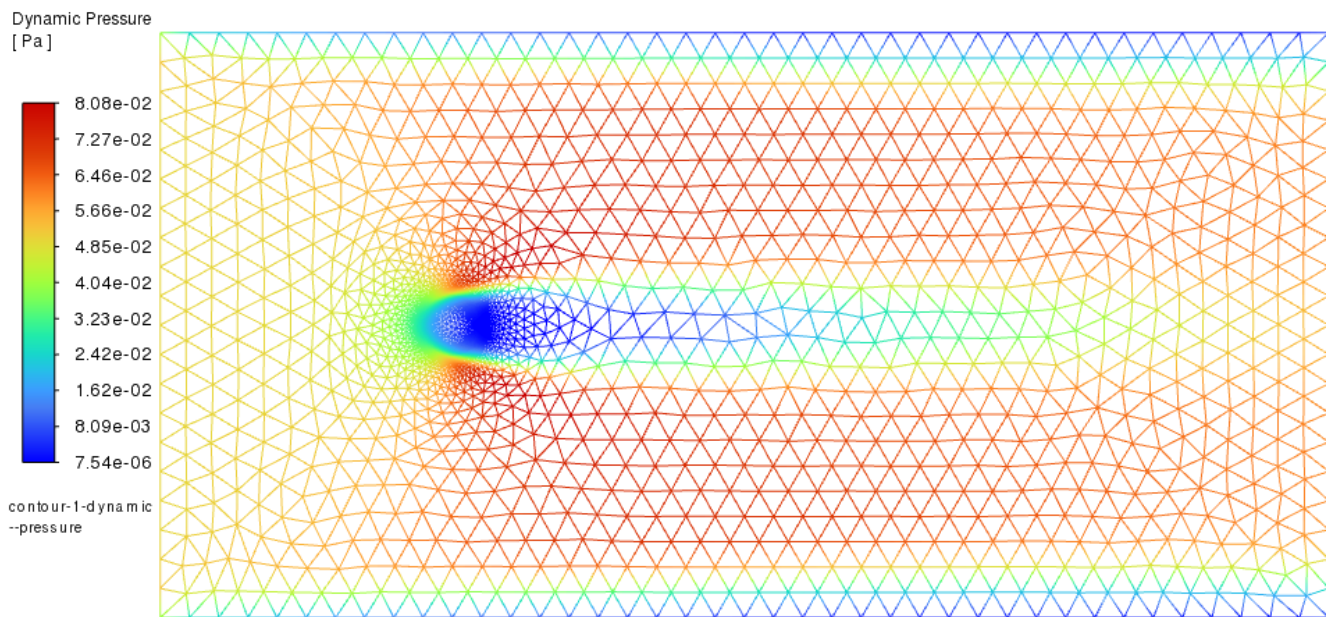
contour-1-x-velocity



contour-1-velocity-magnitude

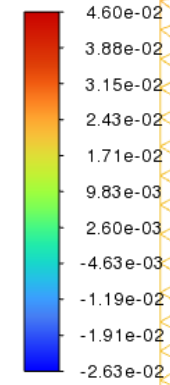


contour-1-dynamic-pressure

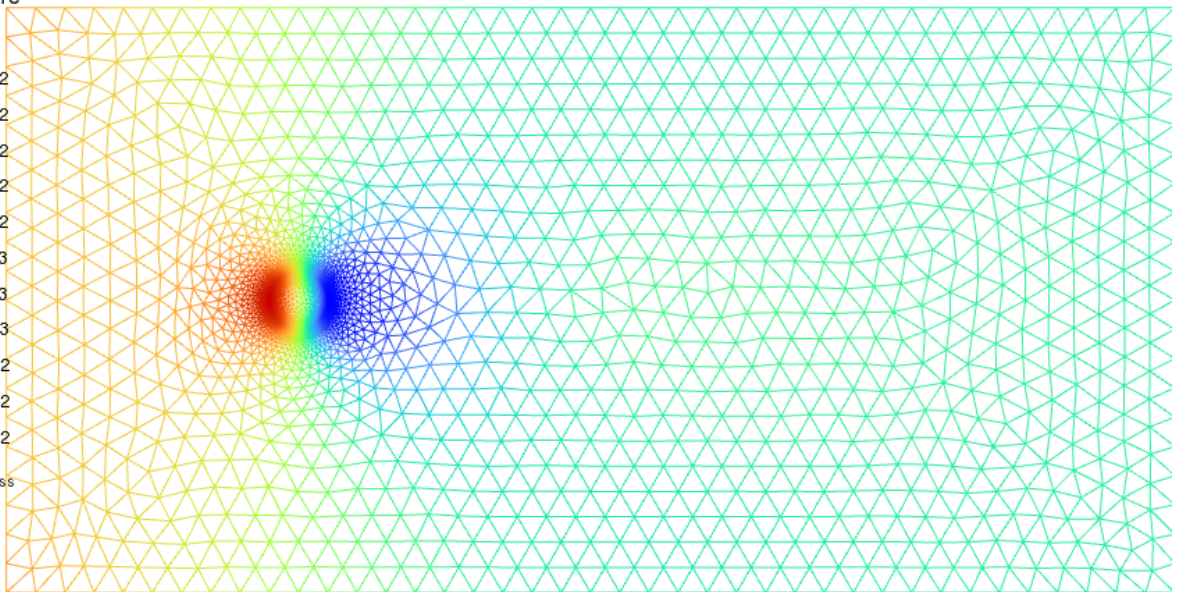


contour-1-static-pressure

Static Pressure
[Pa]



contour-1-static-press
-ure

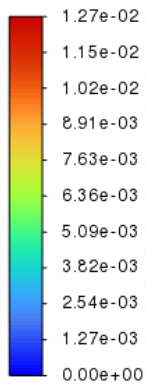


Pathlines

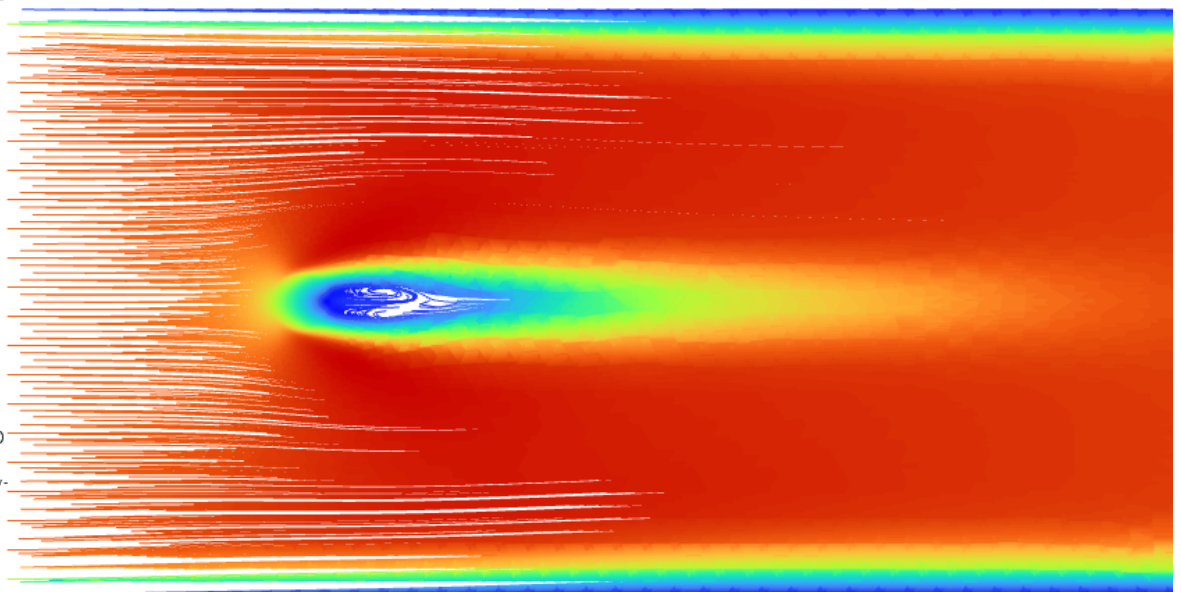
pathlines-1-velocity-magnitude

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Velocity Magnitude
[m/s]

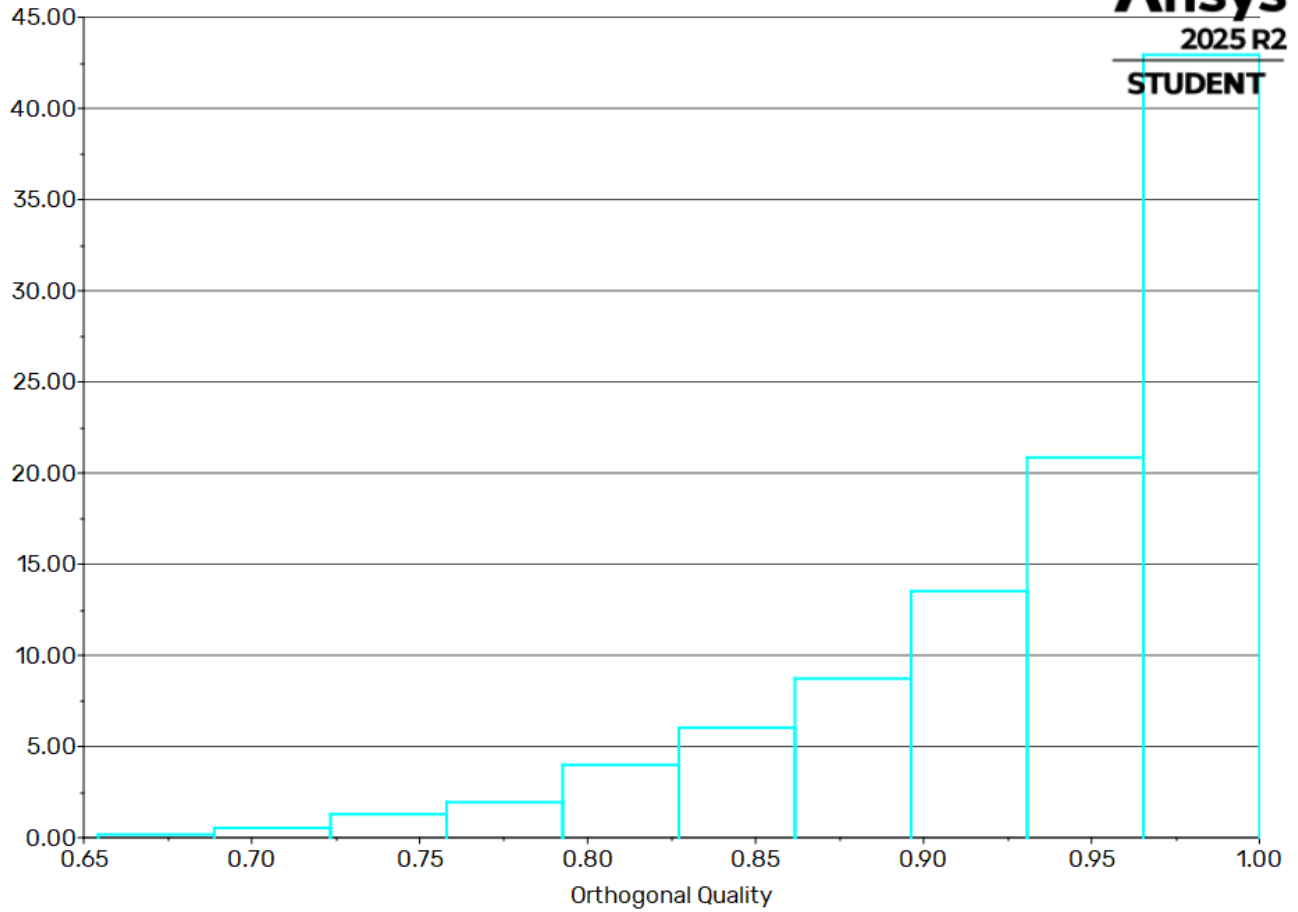


pathlines - 1-velocity-
magnitude

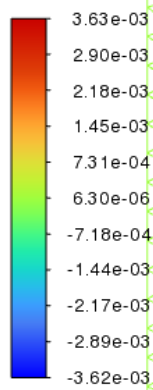


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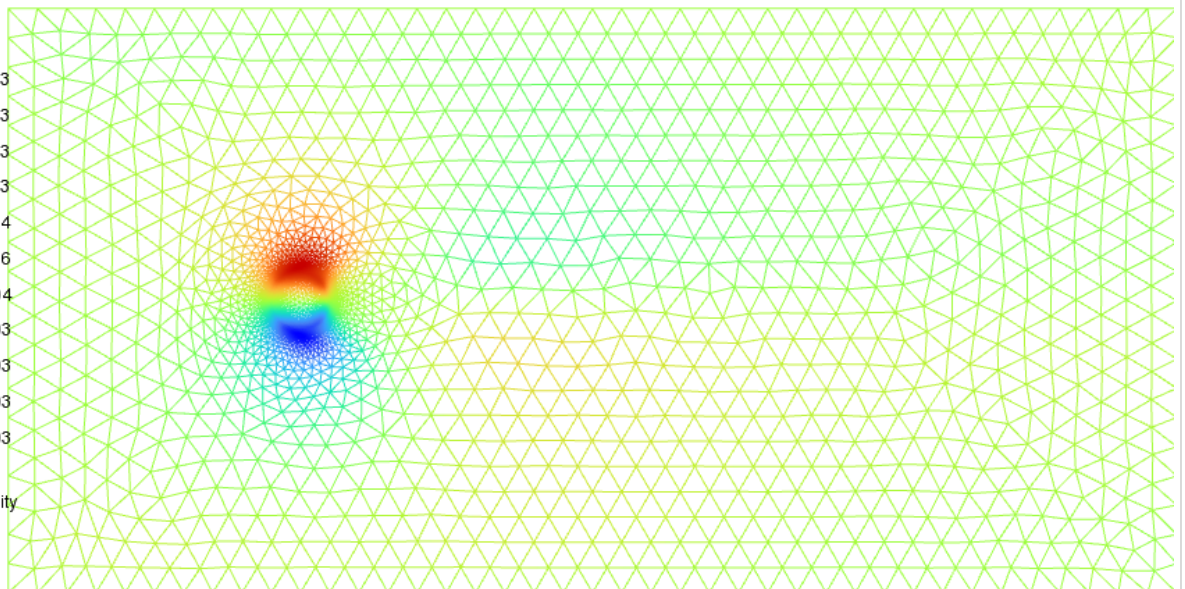
Histogram of Orthogonal Quality (Time=1.5000e+01 s)



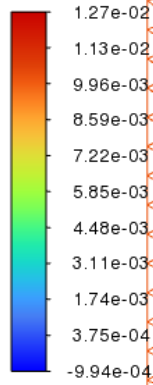
Y Velocity
[m/s]



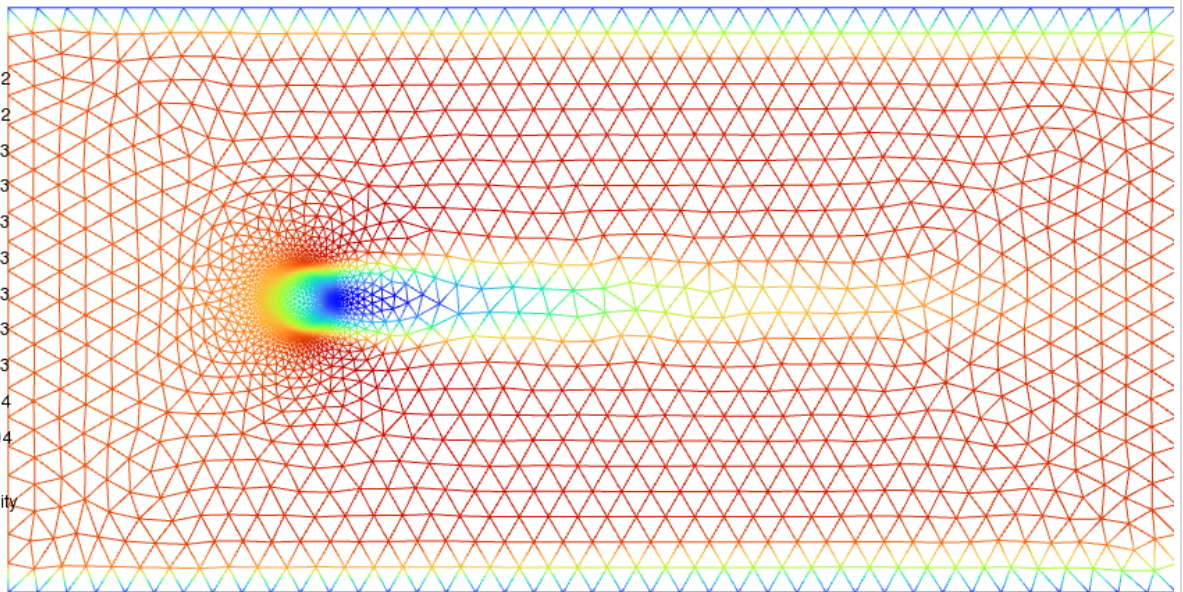
contour-1-y-velocity



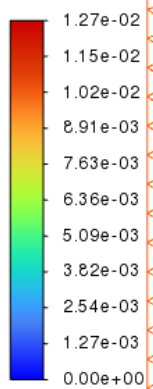
X Velocity
[m/s]



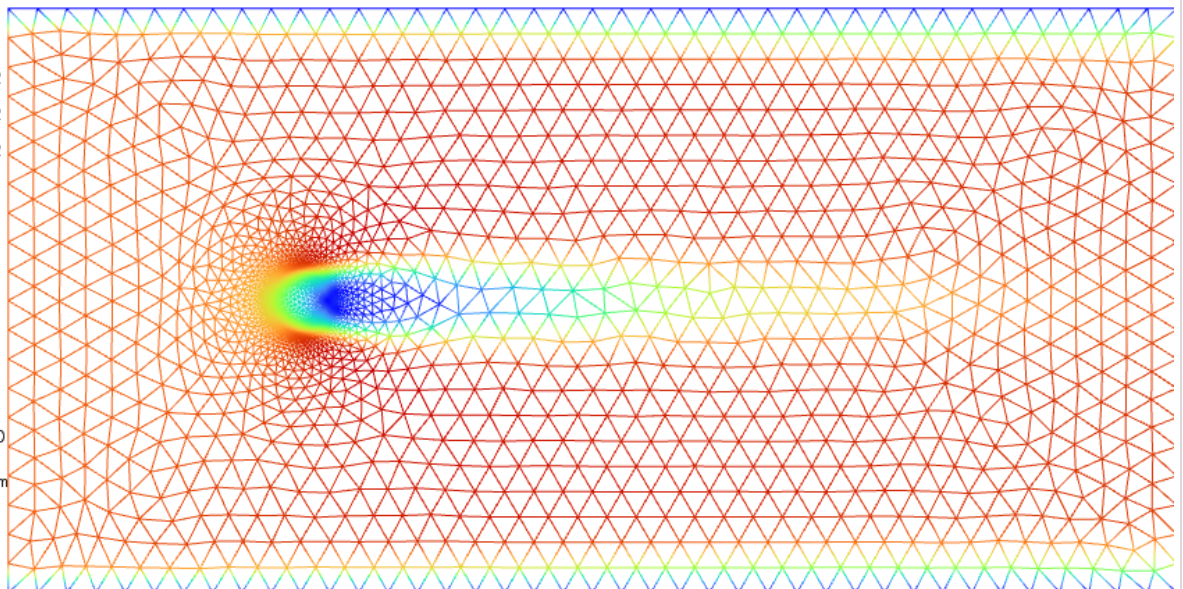
contour-1-x-velocity



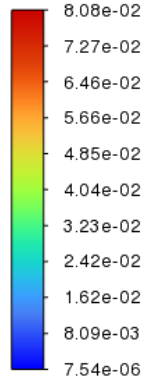
Velocity Magnitude
[m/s]



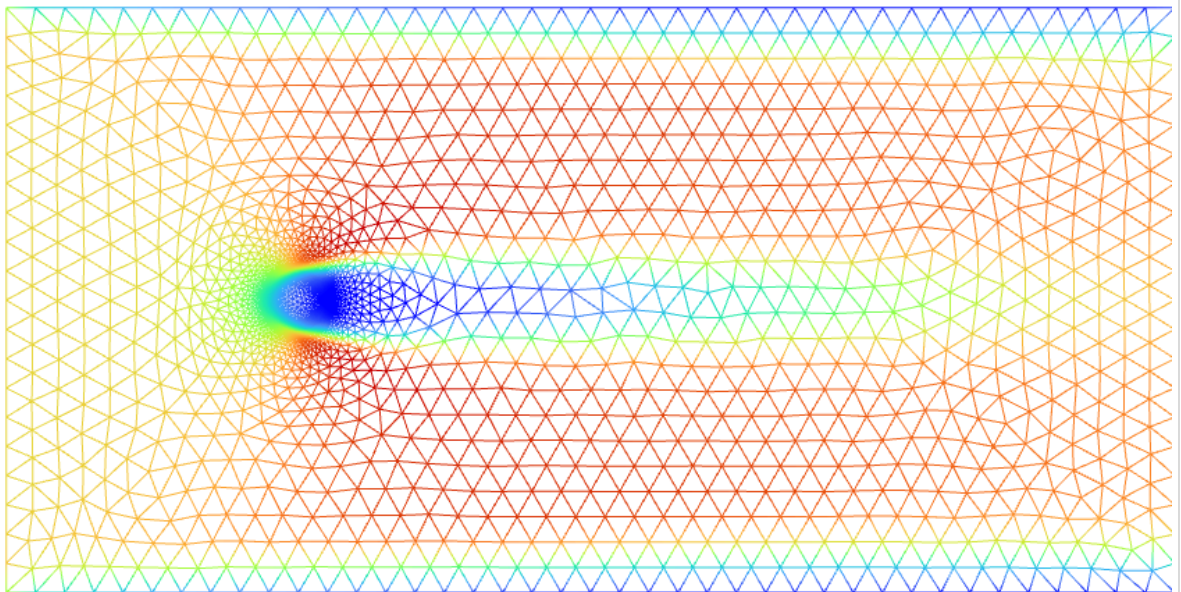
contour-1-velocity-magnitude



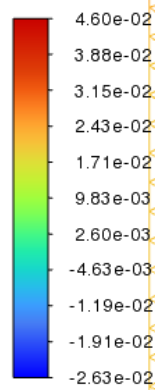
Dynamic Pressure
[Pa]



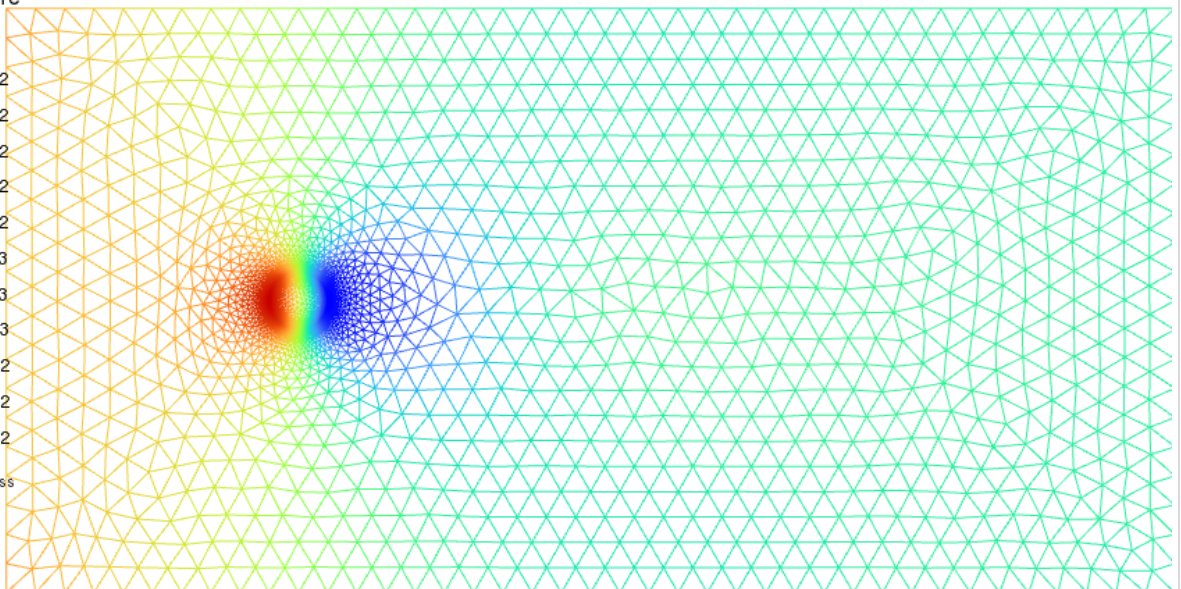
contour-1-dynamic
-pressure



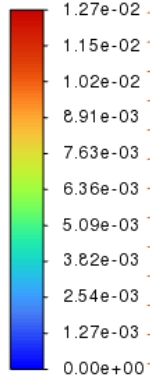
Static Pressure
[Pa]



contour-1-static-press
-ure



Velocity Magnitude
[m/s]



pathlines - 1-velocity-
magnitude

